Social and Technological Networks

2026/2027 1st semester

Lab 1

Introduction

- Many phenomena of everyday life can be described using networks (graphs):
  - *Vertices (nodes):* entities
  - *Edges (links):* connections between the vertices
- Examples:
  - Social networks (e.g., friendships, followers)
  - Transportation networks (roads, public transport)
  - Computer networks
  - Biological networks (e.g., interacting proteins)
  - …

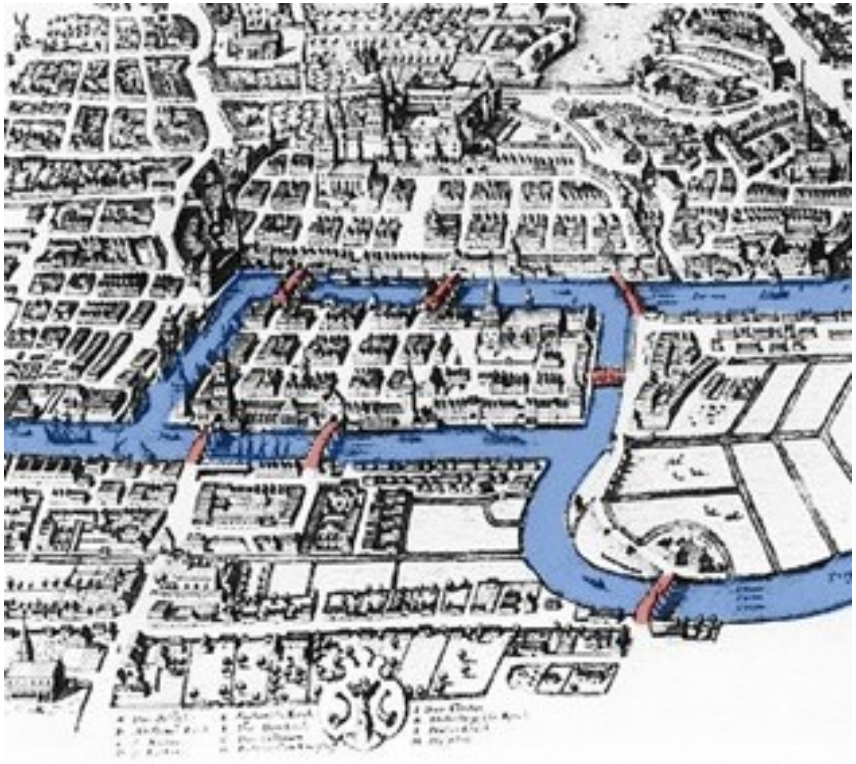
Introduction

Why do we study networks?

- From a data science perspective, networks are the natural model of data relationships.
- They help us understand the structure and dynamics of complex systems.
- Some relevant questions include:
  - Who is the most important actor in the network?
  - What groups or communities emerge?
  - How does information, a disease, or a vehicle travel through the system?

The Bridges of Königsberg problem

- One of the first well-known problems of graph theory.
- The city of Königsberg was divided by the Pregel River.
- The city had 2 islands and 2 riverbanks, connected by 7 bridges.
- The question: Is it possible to traverse the city by crossing each bridge exactly once?



The Bridges of Königsberg problem

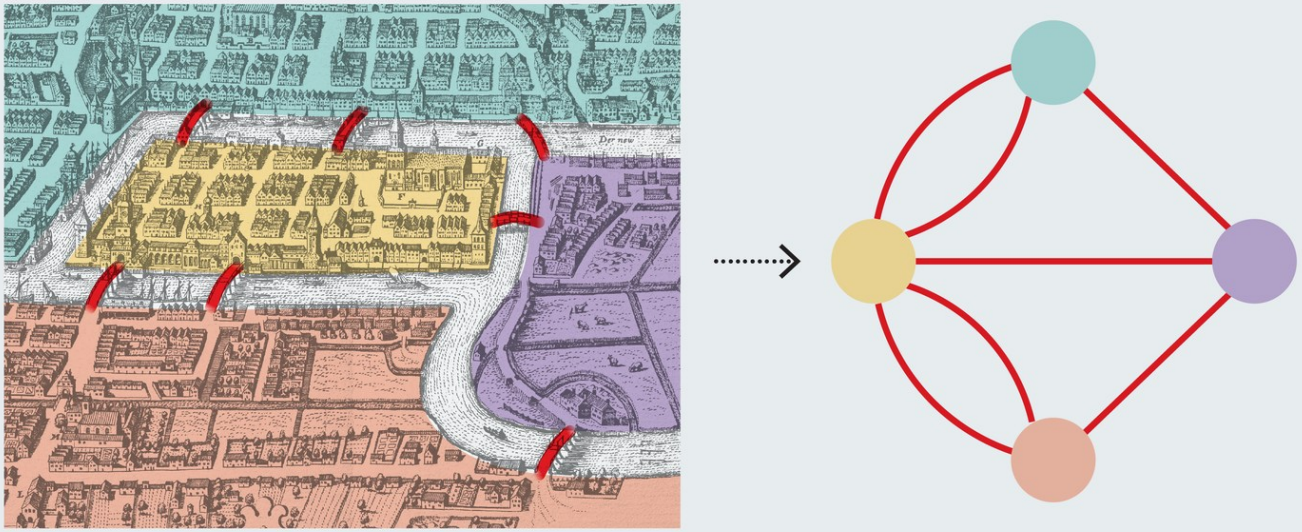


Figure 1: Map of Königsberg with its 7 bridges, and the corresponding graph. Vertices: city districts (islands and riverbanks), edges: bridges.

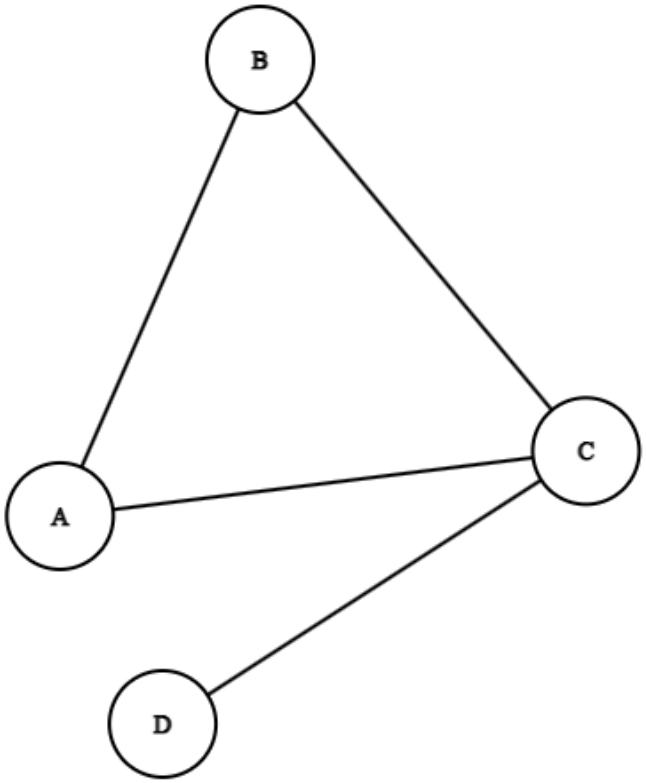
Graphs

- **Graph:**
  - A graph $G(V, E)$r is a structure consisting of a set of vertices V and edges E.
  - Vertices are represented as points, and edges are represented as lines connecting the points.
- **Order of a graph:** the number of vertices.
- **Empty graph:** has one or more vertices but no edges.
- **Trivial graph:** has exactly one vertex and no edges.
- **Degree:** the number of edges incident to a vertex.
- **Adjacency:**
  - Two *vertices* are adjacent if they are connected by an edge.
  - Two *edges* are adjacent if they share a common vertex.

Graphs

Example (simple graph):

- Let $G = (V, E)$ be a graph, where:
  - $V = \{A, B, C, D\}$ and
  - $E = \{(A, B), (A, C), (B, C), (C, D)\}$
- **Order of the graph:** 4
- **Empty:** no, **Trivial:** no
- **Degrees:**
  - $deg(A) = 2$
  - $deg(B) = 2$
  - $deg(C) = 3$
  - $deg(D) = 1$



Graphs – Edges

- **Loop edge:**
  - An edge whose two endpoints coincide, i.e., it connects a vertex to itself.
- **Edge multiplicity:**
  - The number of edges connecting the same pair of vertices.
  - **Simple edge:** There is exactly one edge between two vertices.
  - **Multiple edge:** There are two or more edges between the same pair of vertices.

Graphs – By edges

- **Simple graph:**
  - Contains no loop edges and no multiple edges.
- **Multigraph:**
  - May contain multiple edges (more than one edge between the same two vertices), but no loops.
- **Pseudograph:**
  - May contain both loops and multiple edges.

Graphs – By orientation

- **Undirected graph:**
  - Edges have no direction. They connect two vertices, accessible in both ways.
- **Directed graph (digraph):**
  - Edges are directed, going from a starting vertex to a target vertex.
- **Mixed graph:**
  - Contains both directed and undirected edges.

Exercises – Simple graphs

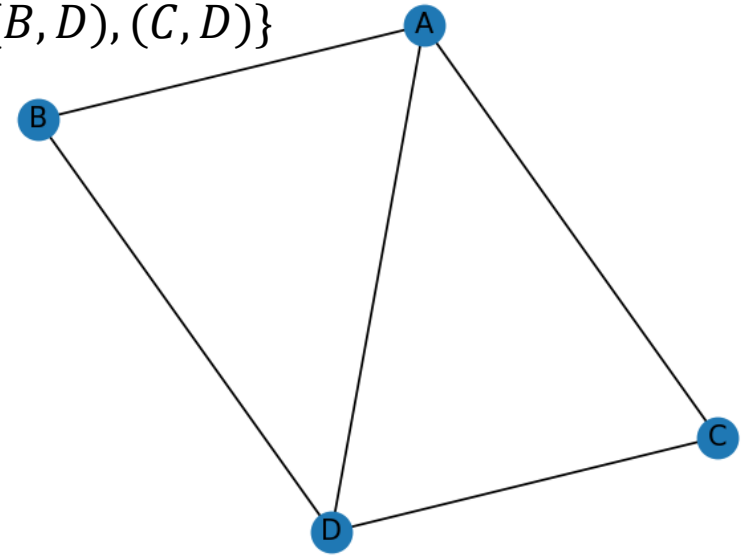
1. In a group of 5 people, can everyone have exactly 3 acquaintances? (The acquaintances are mutual.)
2. How many vertices with odd degrees can there be in a graph?
3. How can we calculate the number of edges in the graph if we know the degree of each vertex?
4. In a round-robin tournament between 8 teams, is it possible that each team has played 1, 2, 2, 2, 4, 5, 7, and 7 matches so far?

Graphs – Walk, path, cycle

- **Walk:**
  - A sequence of vertices and edges, starting and ending at a vertex. Length: The number of edges.
- **Closed walk:**
  - A walk that starts and ends at the same vertex.
- **Path:**
  - A walk in which no vertex is repeated.
- **Cycle:**
  - A closed walk in which no vertex is repeated, except the start/end vertex.

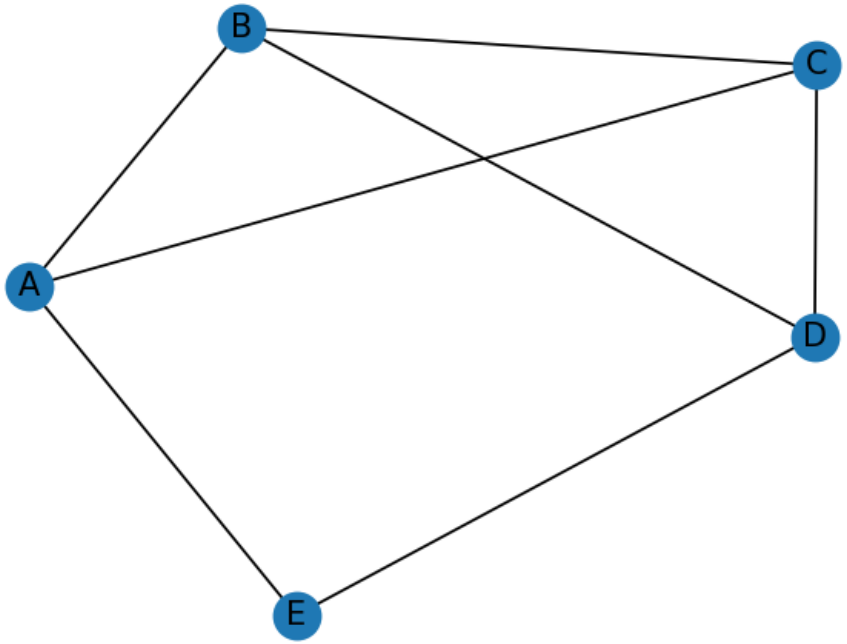
Graphs – Walk, path, cycle

- **Eulerian path:**
  - A path that traverses every edge exactly once.
  - A graph has an Eulerian path if and only if exactly zero or two vertices have odd degree.
- Example:
  - $V = \{A, B, C, D\}$, $E = \{(A, B), (A, C), (A, D), (B, D), (C, D)\}$
  - Degrees:
    - A: 3 (connected to B, C, D),
    - B: 2 (connected to A, D),
    - C: 2 (connected to A, D),
    - D: 3 (connected to A, B, C)
  - One possible Eulerian path:
    - A → B → D → C → A → D



Graphs – Walk, Path, Cycle

- **Hamiltonian path:**
  - A path that visits every vertex exactly once.
- **Example:**
  - $V = \{A, B, C, D, E\}$ and $E = \{(A, B), (B, C), (C, D), (D, E), (E, A), (A, C), (B, D)\}$
  - Adjacency:
    - A: connected to B, C, E
    - B: connected to A, C, D
    - C: connected to A, B, D
    - D: connected to B, C, E
    - E: connected to A, D
  - One possible Hamiltonian path:
    - A → B → C → D → E



The Bridges of Königsberg problem

- **Leonhard Euler (1707–1783)**
  - Swiss mathematician, physicist, astronomer, logician, and engineer.
  - Euler formulated the Bridges of Königsberg problem in a mathematically rigorous way and generalized it.
  - He showed that it is *impossible to traverse* all bridges in the required way.
  - Condition: For such a path (Eulerian path) to exist, at most two vertices can have odd degree.
  - In Königsberg, all vertices had odd degree → no such path exists.

Graphs – Vertices

- **Isolated vertex:**
  - A vertex with no incident edges.
- **Clique:**
  - A set of vertices where every pair is adjacent (a complete subgraph).
- **Independent set:**
  - A set of vertices where no two are adjacent.
- **Bipartite graph:**
  - Vertices can be partitioned into two sets, with edges only between the sets, not within.

Subgraphs

- **Subgraph:**
  - A graph consisting of a subset of the vertices and edges of the original graph.
- **Spanning subgraph:**
  - A subgraph that contains all vertices of the original graph.
- **Induced subgraph:**
  - A subgraph containing all edges between the chosen subset of vertices.
- **Isomorphism:**
  - A mapping between two graphs preserving adjacency between vertices and edges.